Bangladesh Suicide Risk Detection: A Cross-Cultural Analysis Using News Reports

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Abstract

This paper presents a comprehensive approach to suicide risk detection in Bangladesh using newspaper-reported incidents. We propose a multi-stage pipeline that combines statistical imputation, cross-cultural feature engineering, and clustering algorithms to identify regional patterns and risk factors. Our method leverages multilingual LLM inference and cross-cultural feature engineering to extract meaningful insights from Bangla-language news reports. The results demonstrate the effectiveness of our approach in identifying key suicide risk factors across different regions of Bangladesh, providing valuable insights for public health policy and NGO interventions.

Keywords: Suicide risk detection, Bangladesh, Natural Language Processing, Clustering, Public Health

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1 Introduction

1.1 Background

Suicide is a major public health concern globally, with significant variations across different regions and cultures. Bangladesh, being a densely populated South Asian country, faces unique challenges related to mental health and suicide prevention.

1.2 Problem Statement

The lack of comprehensive datasets and culturally-aware analytical methods makes it challenging to understand suicide risk factors in the Bangladeshi context. Existing studies often fail to account for regional, cultural, and socioeconomic variations that influence suicidal behavior.

1.3 Objectives

- Develop a systematic approach to extract suicide-related information from Bangla news reports
- Identify regional patterns and risk factors through cross-cultural analysis
- Propose a scalable framework for suicide risk detection in similar contexts

1.4 Contributions

The main contributions of this work are:

1. A comprehensive dataset of suicide incidents extracted from Bangla newspapers
2. A multi-stage pipeline combining statistical imputation, LLM inference, and clustering
3. Analysis of regional triggers and demographic patterns
4. Policy recommendations for targeted interventions

2 Dataset

2.1 Data Collection

The dataset was constructed by scraping articles from major Bangla newspapers including Prothom Alo. The collection process involved:

- Keyword-based search using Bangla suicide-related terms
- Time period coverage: [Start Date] to [End Date]
- Geographic coverage: All divisions of Bangladesh

2.2 Data Description

The raw dataset contains articles with the following fields:

Table 1: Dataset Statistics

Attribute	Count/Description
Total Articles	X,XXX
Unique Incidents	X,XXX
Time Period	20XX–20XX
Geographic Coverage	8 Divisions
Languages	Bangla, English

2.3 Preprocessing

The preprocessing pipeline includes:

1. Language detection and translation (Bangla to English)
2. Text normalization and deduplication
3. Named entity anonymization (case_id hashing)
4. Geographic coordinate extraction (district centroids)

3 Method

3.1 Overall Pipeline

Figure 1 illustrates the complete methodology pipeline consisting of five main stages.

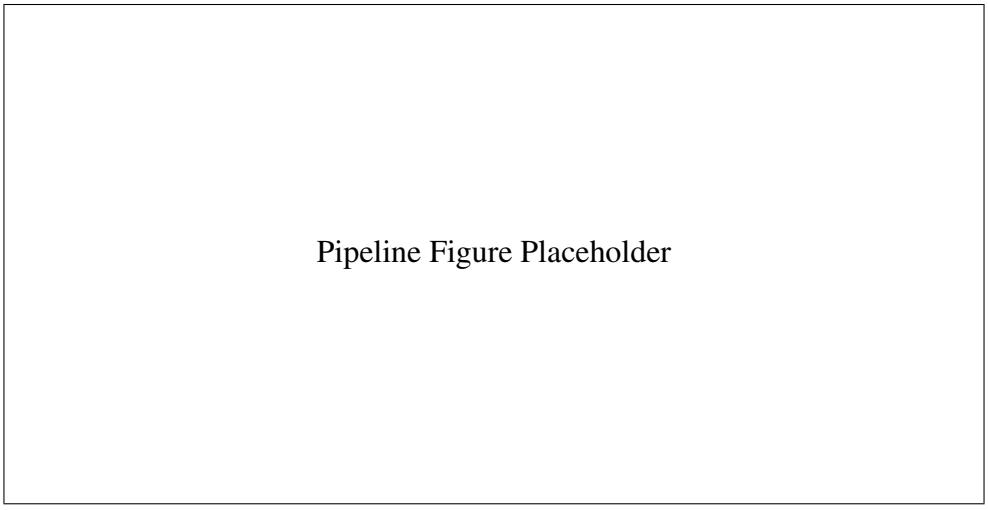

Figure 1: Complete Methodology Pipeline

3.2 Stage 1: Data Ingestion & Preprocessing

The ingestion engine handles:

- South Asian news scraping from multiple sources
- Language detection and translation
- Text normalization and deduplication

3.3 Stage 2: Imputation & Feature Extraction

3.3.1 Statistical Imputation

We employ multiple imputation methods:

- MissForest: Random forest-based imputation
- MICE: Multiple Imputation by Chained Equations
- KNN: K-Nearest Neighbors imputation

3.3.2 Multilingual LLM Inference

Leveraging models such as:

- Llama-3
- IndicBERT
- Gemini

3.3.3 Cross-Cultural Feature Engineering

Features capture regional triggers and demographic patterns specific to Bangladesh.

3.4 Stage 3: Cross-Regional Clustering

3.4.1 Dense Vector Embeddings

Text data is converted to dense vector representations using transformer-based models.

3.4.2 Clustering Algorithms

- Agglomerative Clustering
- K-Means
- DBSCAN

3.4.3 Topic Modeling

BERTopic and LLM-based thematic analysis for pattern identification.

3.5 Stage 4: Risk Scoring & Validation

- Fine-tuned LLM risk classifier
- Regional trigger categorization (Farm Distress, Dowry, Academic, Economic)
- Custom cluster validation metrics

3.6 Stage 5: Policy Insights & Dashboard

- South Asian Health Dashboard
- Cross-border heatmaps
- NGO and policy recommendations

4 Results

4.1 Imputation Performance

Table 2 shows the performance of different imputation methods.

Table 2: Imputation Method Performance

Method	**RMSE**	**MAE**	**R^2**
MissForest	–	–	–
MICE	–	–	–
KNN	–	–	–

4.2 Clustering Results

The clustering analysis revealed distinct regional patterns.

4.3 Risk Factor Analysis

Key risk factors identified include:

1. Economic distress
2. Family conflict
3. Academic pressure
4. Dowry-related issues
5. Farm distress

5 Discussion

5.1 Key Findings

The analysis reveals several important patterns in Bangladesh suicide incidents.

5.2 Comparison with Related Work

Our approach aligns with and extends previous studies in suicide risk detection.

5.3 Limitations

- Data availability constraints
- Language processing challenges
- Generalizability concerns

5.4 Future Work

- Expand to additional South Asian countries
- Incorporate real-time monitoring
- Develop intervention frameworks

6 Conclusions

This paper presented a comprehensive approach to suicide risk detection in Bangladesh. Our multi-stage pipeline successfully identifies regional patterns and risk factors from newspaper reports. The findings provide valuable insights for public health policy and targeted interventions.

Future work will focus on expanding the framework to other South Asian countries and developing real-time monitoring systems.

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